Chapter 3

ANALYTICAL GEOMETRY

Learning Objectives

After studying this chapter, the students will be able to understand

- the locus
- the angle between two lines.
- the concept of concurrent lines.
- the pair of straight lines
- the general equation and parametric equation of a circle.
- the centre and radius of the general equation of a circle.
- the equation of a circle when the extremities of a diameter are given.
- the equation of a tangent to the circle at a given point.
- identification of conics.
- the standard equation of a parabola, its focus, directrix, latus rectum and commercial applications



Rene Descartes
(1596-1650)

Introduction

The word "Geometry" is derived from the word "geo" meaning "earth" and "metron" meaning "measuring". The need of measuring land is the origin of geometry.

"Geometry" is the study of points, lines, curves, surface etc., and their properties. The importance of analytical geometry is that it establishes a correspondence between geometric curves and algebraic equations.

A systematic study of geometry by the use of algebra was first carried out by celebrated French Philosopher and mathematician Rene Descartes (1596-1650), in his book' La Geometry', published in 1637. The resulting combination of analysis and geometry is referred now as analytical geometry. He is known as the father of Analytical geometry

Analytical geometry is extremely useful in the aircraft industry, especially when dealing with the shape of an airplane fuselage.

3.1 Locus

Definition 3.1

The path traced by a moving point under some specified geometrical condition is called its locus.

3.1.1 Equation of a locus

Any relation in x and y which is satisfied by every point on the locus is called the equation of the locus.

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